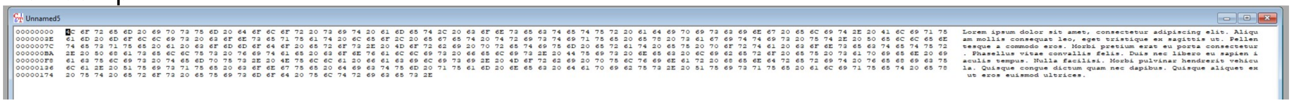
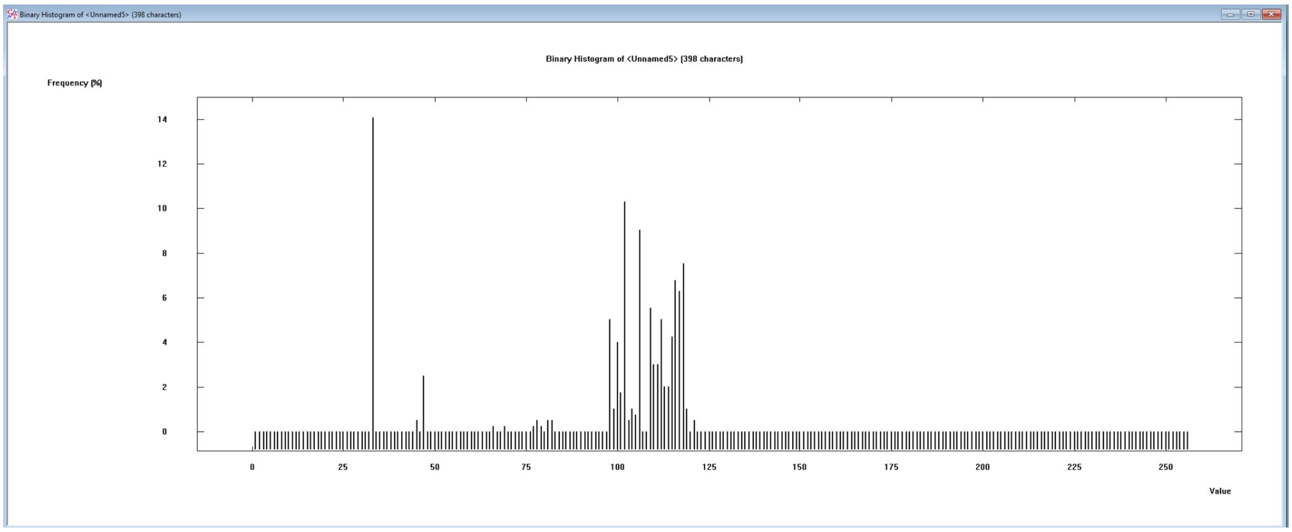
Adam Cherek s207250

1.

Plain text used: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam mollis consequat leo, eget tristique ex sagittis ut. Pellentesque a commodo eros. Morbi pretium erat eu porta consectetur. Phasellus vitae convallis felis. Duis nec libero eu sapien iaculis tempus. Nulla facilisi. Morbi pulvinar hendrerit vehicula. Quisque congue dictum quam nec dapibus. Quisque aliquet ex ut eros euismod ultrices.

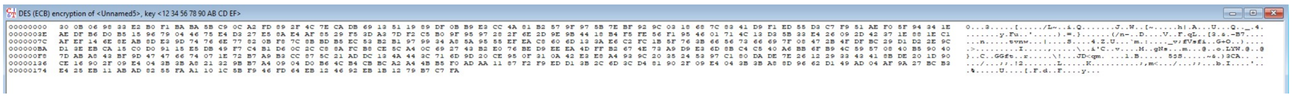
Hex dump:

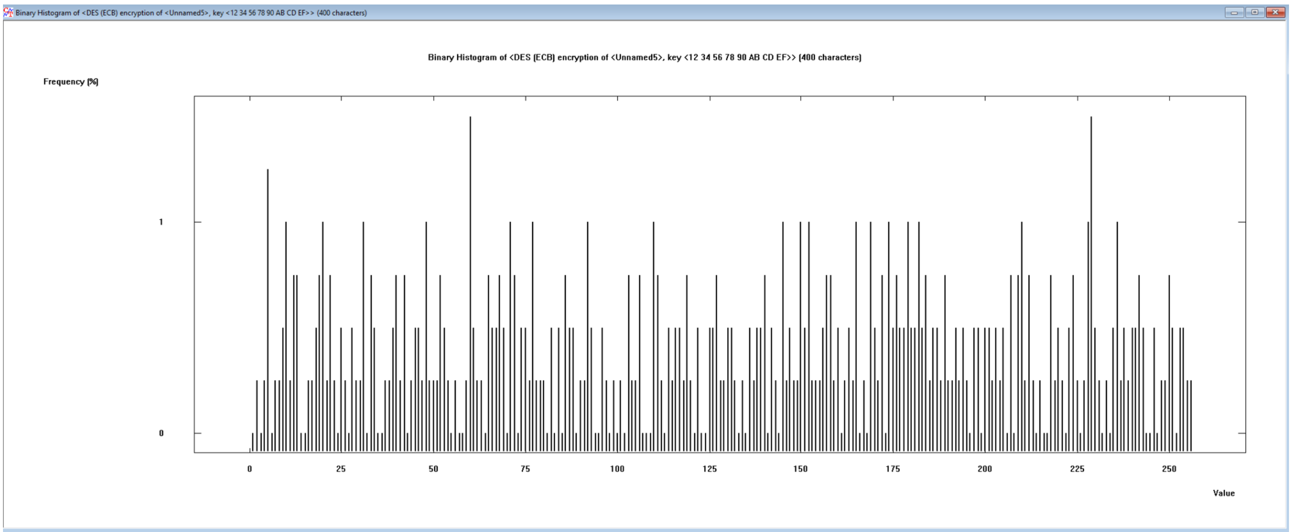




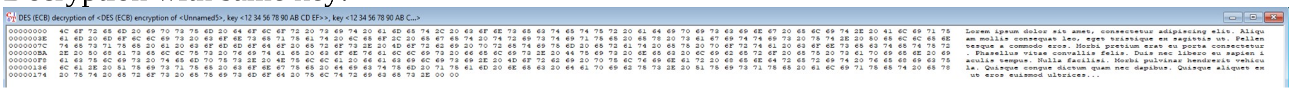
Key: 1234567890ABCDEF

Encrypted:





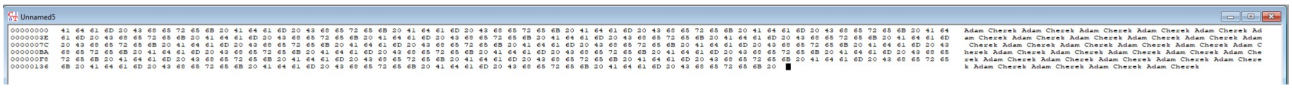
Decryption with same key:



Conslusion:
The encrypted text have much better characters distribution.

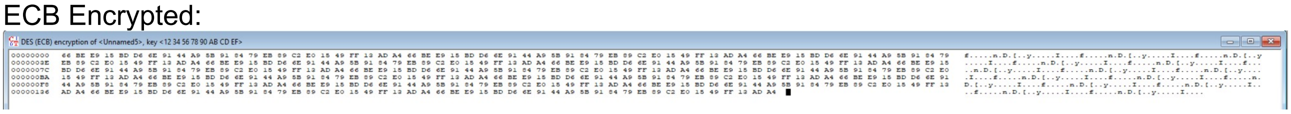
2.

Plain text: Adam Cherek



Key: 1234567890ABCDEF

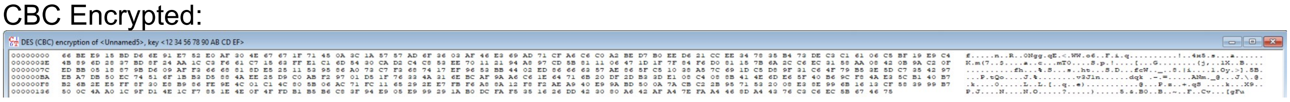
ECB Encrypted:



Periodicity:

No.	Offset	Length	Number of c...	Cycle content	Cycle content (hex)
1	1	24	15	fïé.˝Ön.D©[..yë.Åf.I	66 BE E9 15 BD D6 6E 91 44 A9 5B 91 84

CBC Encrypted:



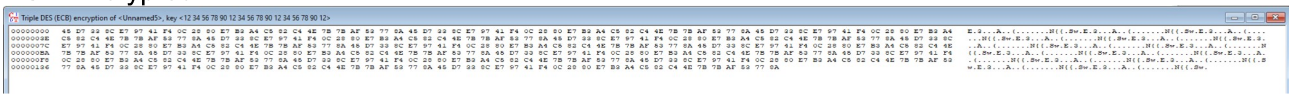
Periodicity: No periodicity found

3.

Plain text: Same as the one in task 2

Key: 12345678901234567890123456789012

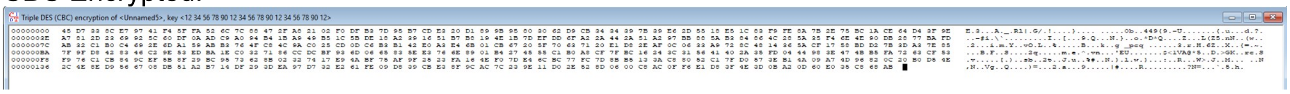
ECB Encrypted:



Periodicity:

No.	Offset	Length	Number of c...	Cycle content	Cycle content (hex)
1	1	24	15	E.3.ç.Aô.(.ç¦¤L.ÄN{{	45 D7 33 8C E7 97 41 F4 0C 28 80 E7 B3 A

CBC Encrypted:



Periodicity: No periodicity found

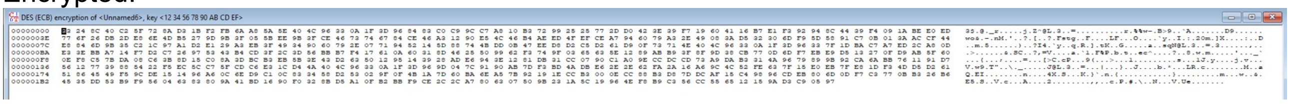
Conclusions: 3DES = DES Encrypt with K1, DES Decrypt with K2 and again DES Encrypt with K3 (if I remember correctly) so to make 3DES transmitter + DES Receiver work, all keys K must be identical K1 = K2 = K3

4.

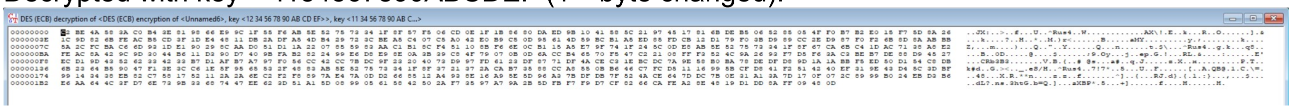
Plain text: Store leftover pancakes in an airtight container in the fridge for about a week. Refrain from adding toppings (such as syrup) until right before you serve them so the pancakes don't get soggy.
To reheat pancakes, you have a few options: warm a batch in a 350 degrees F oven (covered) for 8 to 10 minutes, microwave individual pancakes in 20 to 30 second bursts with a damp paper towel, or reheat them in a skillet over medium-low heat for 30–60 seconds per side until warmed through.

Key: 1234567890ABCDEF

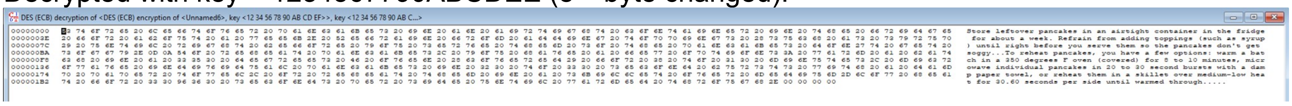
Encrypted:



Decrypted with key – 1134567890ABCDEF (1st byte changed):



Decrypted with key – 1234567790ABCDEE (8th byte changed):

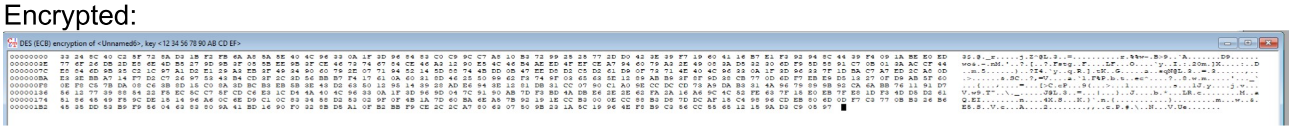


Conclusion: Changing 8th bit makes the decryption readable, but changing other bits not. Its an avalanche effect.

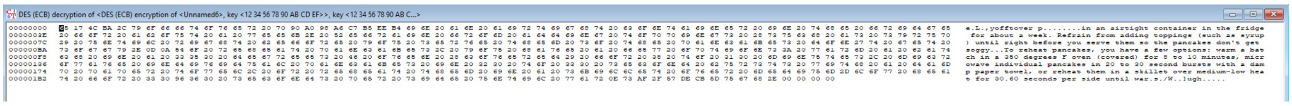
5.

Plain text: Same as one in task 4
Key: 1234567890ABCDEF

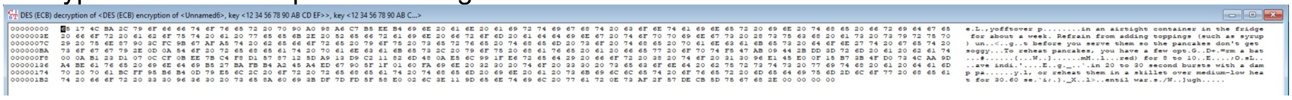
Encrypted:



Decrypted after 3 characters changed:



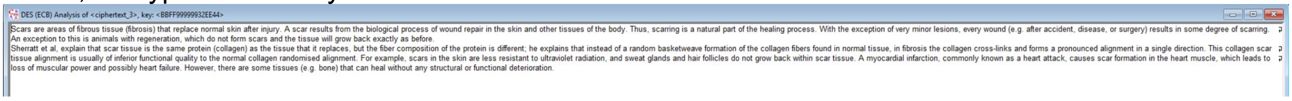
Decrypted after few sequences changed:



Conclusions: Text remains very readable after changing few characters, and still somehow readable after changing multiple sequences.

6.

Text 3, decryption with 5 bytes: about 9 seconds



Text 3, decryption with 4 bytes: about 22 minutes

Text 3, decryption with 3 bytes: about 1.9 days

7.

Advantages of DES algorithm:

1. DES has been around a long time (since 1977), even no actual weaknesses have been discovered and the most effective attack is still brute force.
2. DES is an official United States Government standard. The Government is needed to re-certify, DES every five years and ask it be restored if essential.
3. DES is also an ANSI and ISO standard. Because DES was designed to run on 1977 hardware, it is rapid in hardware and associatively quick in software.
4. It supports functionality to save a file in an encrypted format which can only be accessed by supporting the correct password.
5. It can change the system to create the directories password protected.
6. It can review a short history of DES and represent the basic structures.
7. It can define the building block component of DES.
8. It can define the round keys generation process and to interpret data encryption standard.
9. It can provide that private information is not accessed by other users.
10. Some users can use the similar system and still can work individually.

Disadvantages of DES algorithm:

1. The 56 bit key size is the largest defect of DES and the chips to implement one million of DES encrypt or decrypt operations a second are applicable (in 1993).
2. Hardware implementations of DES are very quick.
3. DES was not designed for application and therefore it runs relatively slowly.
4. In a new technology, it is improving a several possibility to divide the encrypted code, therefore AES is preferred than DES.